

EGB240 Electronic Design

Assessment 1: PCB Alarm Circuit Design Portfolio

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Executive Summary

The purpose of this document is to display the design process for a PCB dual-tone siren circuit implemented using the following specifications:

- 3v input from two AA batteries
- 30mA Supply
- Slide Switch for activation
- Alternating between 242Hz and 2424Hz at 1.5Hz Interval
- board size equates to 53.41 x 37.30 x 12 mm (excludes battery holder)

To reduce size, the siren is constructed on a double-sided PCB. Every capacitor used in this design is electrolytic to avoid the occasional unstable nature of that of a ceramic capacitor and ensure the stability of the low frequency oscillator.

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1. Circuit schematic

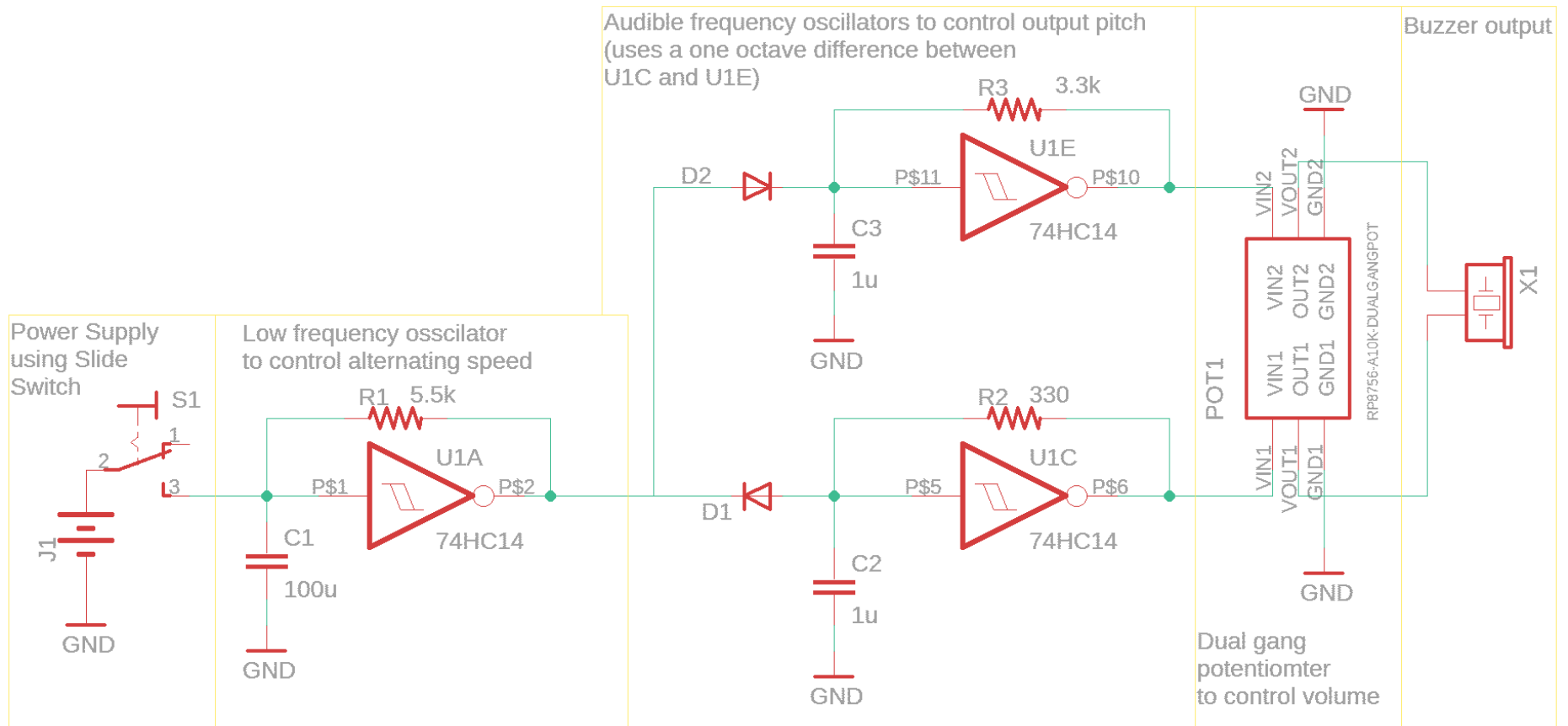


Figure 1: PCB alarm Schematic

2. Summary of design and operation

Shown in (fig. 1) this design features three Schmitt inverters taken from a hex-inverter IC. Each of the used inverters corresponds to one oscillator with 2 audible (242Hz and 2424Hz) and one inaudible (1.5Hz in this case). While creating this design the purpose in mind was that of something like an alarm clock to assist in time management. This design features a dual gang logarithmic potentiometer to simultaneously control the volume of each oscillator such that they are both adjustable and equivalent. This allows for use in different areas such as at home where you can be louder or even in the library where you would need it to be much quieter.

While deciding the audible oscillators many different versions were tested. The final versions of the audible frequencies were chosen due to them being exactly one octave apart. This allows for a less stressful alarm. The low frequency was also chosen for a similar reason. With a 1.5Hz pulse this allows for a much slower alarm meant to inform someone that time is up as opposed to alert them to an emergency for example. Using these frequencies the following component values were obtained.

$$f = \frac{1}{k * RC}$$

The k-value is given as 1.25 and the capacitor for high frequencies is assumed to be 1uF

$$242 = \frac{1}{1.25 * R(1uF)}; R = 330\Omega$$

$$2424 = \frac{1}{1.25 * R(1uF)}; R = 3.3k\Omega$$

For low frequency capacitor is assumed to be 100uF

$$1.5 = \frac{1}{1.25 * R(100uF)}; R = 5.5k\Omega$$

With these component values selected it was then possible to prototype a design. My design, shown in section 1, uses diodes to direct the pulse generated by the low frequency oscillator to either U1E or U1C in an alternating fashion. Each of these Schmitt triggers use basic oscillator principles to then create a pwm waveform with corresponding frequencies to that of the chosen specifications. Once this pulse is outputted but the trigger it is then fed into a dual gang logarithmic potentiometer to control the desired intensity of the wave and thus control the volume.

A logarithmic potentiometer was chosen because of its common use in volume control due to its exact nature of each dial position corresponding to one distinct resistance level. This can be seen in the figure 2.

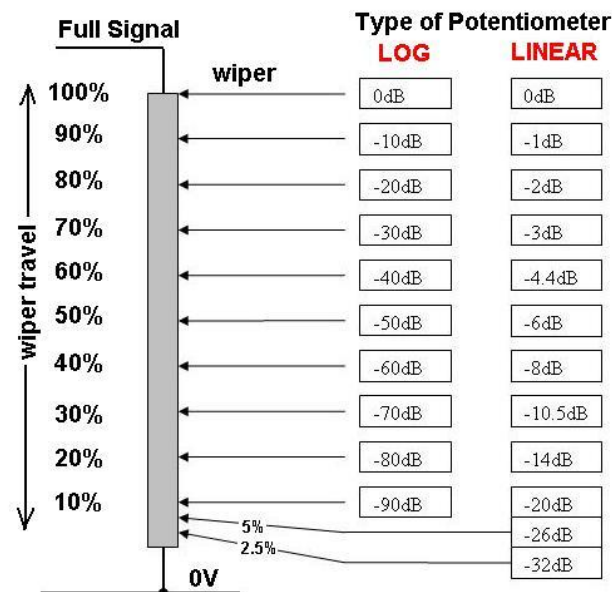
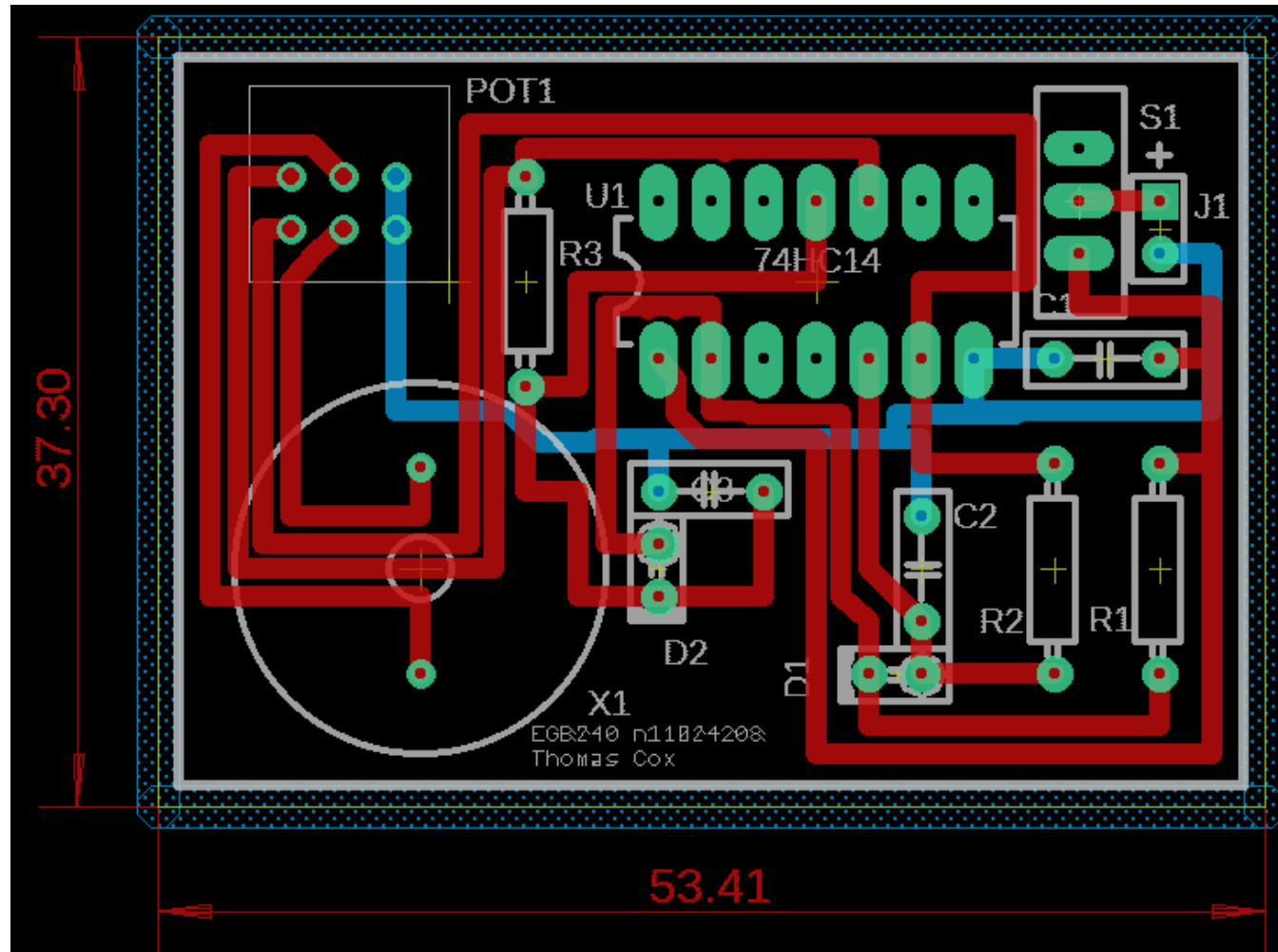


Figure 2: Log vs Lin for audio purposes

This figure shows that when observing the volume output, log potentiometers are much more appropriate due to their increased control. When examining the log wiper position you can see that the db decrease is inversely proportional to the wiper travel. For example at 70% wiper level, db decrease is $= 100 - 70 = 30\text{dB}$ decrease and the same is true for the 40% wiper level which shows a 60dB decrease while the linear model has a much less consistent pattern with -3dB at 70% and -8 at 40% with lower levels such as those in the 0-10% wiper range having a much higher impact on the volume.

3. PCB layout



4. Bill of materials

Designator	Value	Description	Qty	Footprint
J1	3V	Battery holder, 3V, 2xAA, flying leads	1	BATT-3V
S1		Sliding SPDT switch	1	SWITCH-SPDT
C1	100μF	Capacitor, electrolytic radial, 100μF, 16V	1	CAP-DISC-P5.08
C2	1μF	Capacitor, electrolytic radial, 1μF, 63V	1	CAP-DISC-P5.08
C3	1μF	Capacitor, electrolytic radial, 1μF, 63V	1	CAP-DISC-P5.08
R1	5.5k	Resistor, axial, metal-film, 5.5kΩ, 1%, 0.5W	1	AXIAL-P10.16
R2	330k	Resistor, axial, metal-film, 330kΩ, 1%, 0.5W	1	AXIAL-P10.16
R3	3.3k	Resistor, axial, metal-film, 3.3kΩ, 1%, 0.5W	1	AXIAL-P10.16
D1	1N4148	Diode, 100V 150mA, DO-35	2	DO35-P2.54
D2	1N4148	Diode, 100V 150mA, DO-35	2	DO35-P2.54
POT1	10k	Variable Resistor, Double Gang, logarithmic, 10kΩ	1	RP8756-A10K-DUALGANGPOT ¹
X1		Piezo Buzzer	1	PS1720P02
U1	74HC14	IC, Hex inverting Schmitt trigger, DIP-14	1	DIP-14
U1		IC Socket, DIP-14	1	DIP-14

¹ This is a custom footprint. The script for it can be seen in appendix A

Designator	Manufacturer	MPN	Supplier	SKU	MOQ	Price
J1	Generic	Battery holder, 3V, 2xAA, flying leads	Jaycar	PH9202	1	\$0.95
S1	NKK Switches	Sliding SPDT switch	Digikey	360-2922-ND	1	\$3.42
C1	Generic	Capacitor, electrolytic-radial, 100µF, 16V	Jaycar	RE6130	2	\$0.18
C2	Generic	Capacitor, electrolytic-radial, 1µF, 63V	Jaycar	RE6032	2	\$0.15
C3	Generic	Capacitor, electrolytic-radial, 1µF, 63V	Jaycar	RE6032	2	\$0.15
R1	Generic	Resistor, axial, metal-film, 5.5kΩ, 1%, 0.5W	Jaycar	RR0593	8	\$0.11
R2	Generic	Resistor, axial, metal-film, 330kΩ, 1%, 0.5W	Jaycar	RR0560	8	\$0.11
R3	Generic	Resistor, axial, metal-film, 3.3kΩ, 1%, 0.5W	Jaycar	RR0584	8	\$0.11
D1	NXP	1N4148	Jaycar	ZR1100	5	\$0.19
D2	NXP	1N4148	Jaycar	ZR1100	5	\$0.19
POT1	Generic	Variable Resistor, Double Gang, logarithmic, 10kΩ, 0.02W	Jaycar	RP8756	1	\$6.95
X1	TDK	PS1720P02	Element14	1669968	1	\$0.95
U1	Texas Instruments	SN74HC14N	Jaycar	ZC-4821	1	\$1.15
U1	Generic	IC Socket, DIP-14	Jaycar	PI-6501	1	\$0.35

5. Assembly overlay

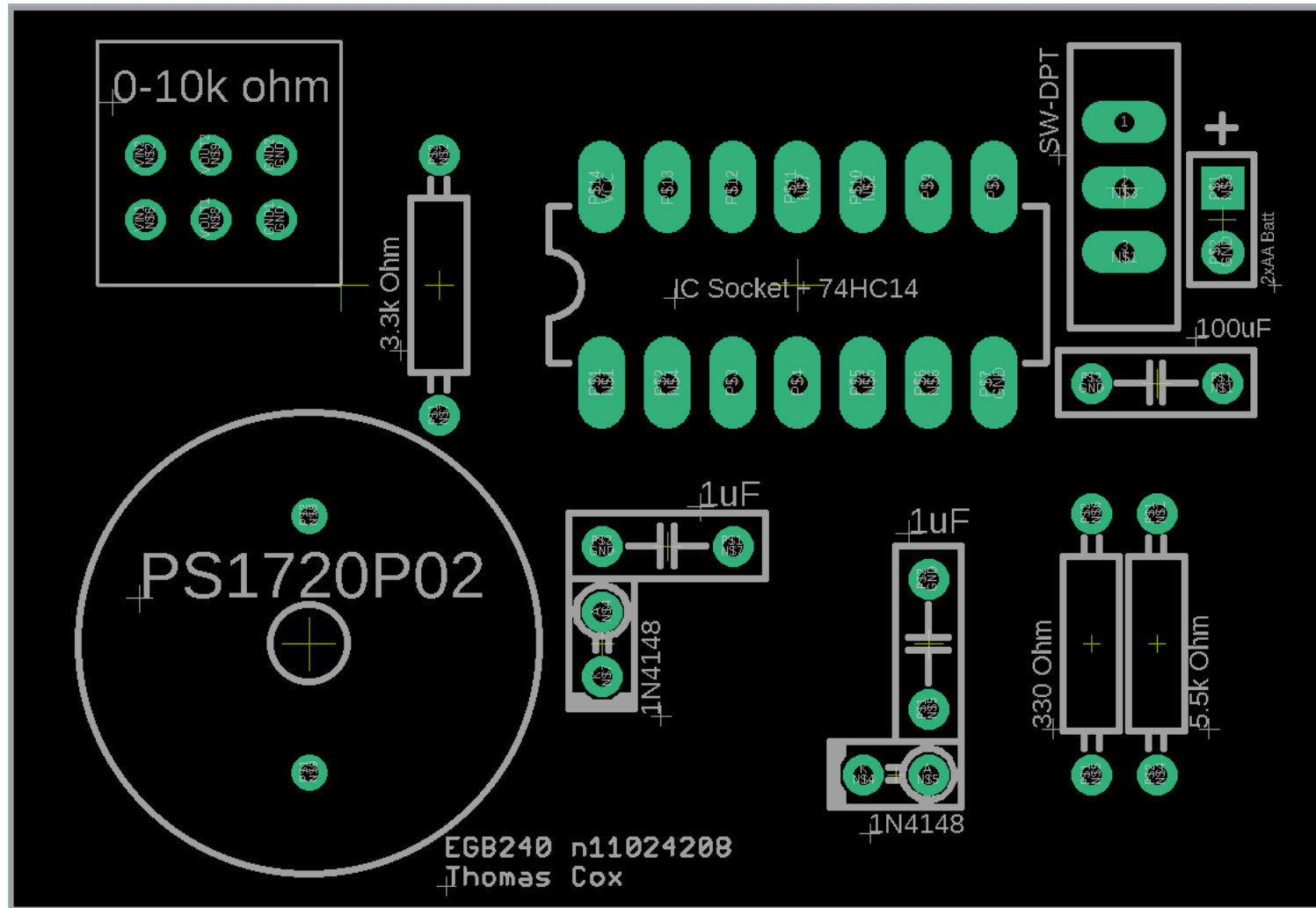
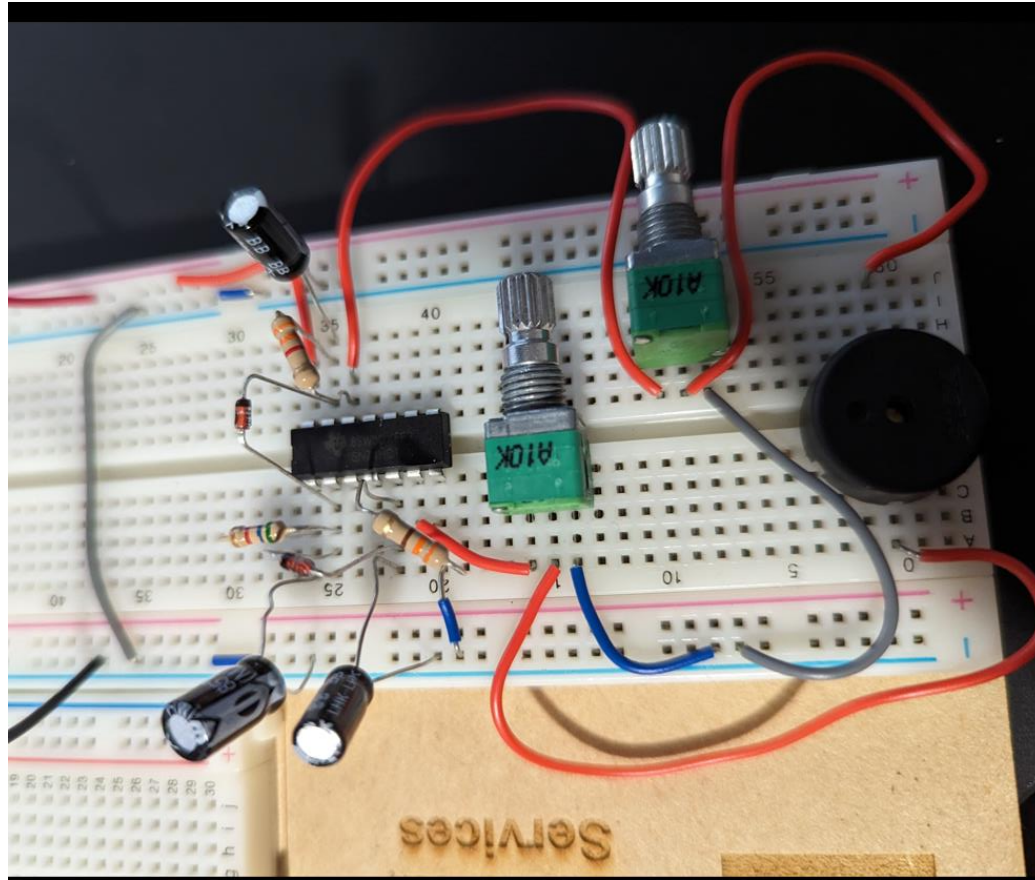


Figure 3: assembly overlay showing each special part and the value for each generic part

6. Photos of assembled prototype



It is important to note that for the purpose of prototyping 2 identical double gang potentiometers were used due to the pins being too close and not being able to bridge the gap in the center. This is solved by the PCB print and only one potentiometer is used in the final design

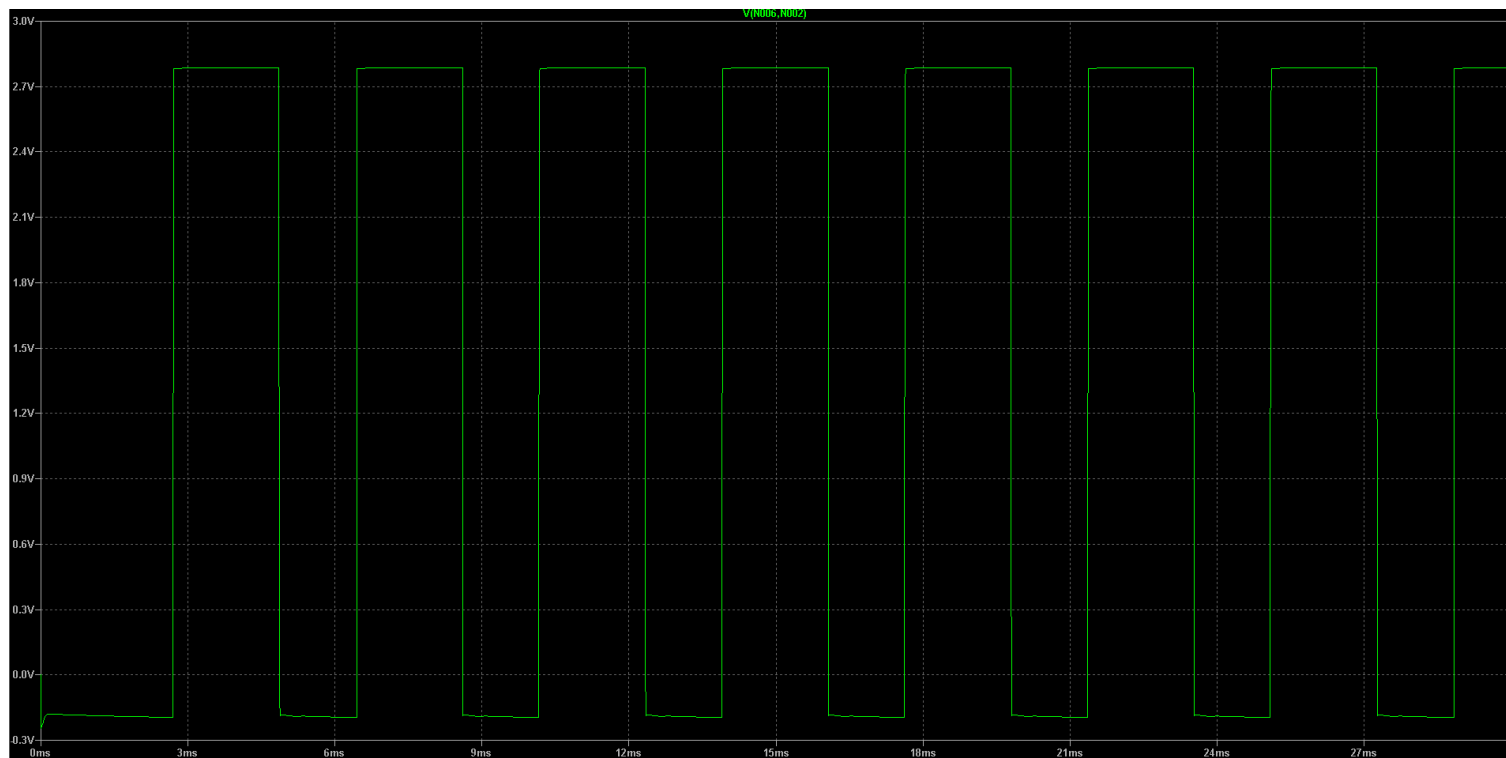
7. Simulation circuit and results

Using the schematic shown in section 1, the following simulation was performed.

Simulation – Voltage across the buzzer

```
.tran 30ms uic
```

```
.prob V(N006,N002)
```



8. Experimental results

Using the prototype shown in section 6, the above simulation was then tested using a digital oscilloscope with the results as follows.



Figure 4: Experimental Results with the potentiometer set to 0%

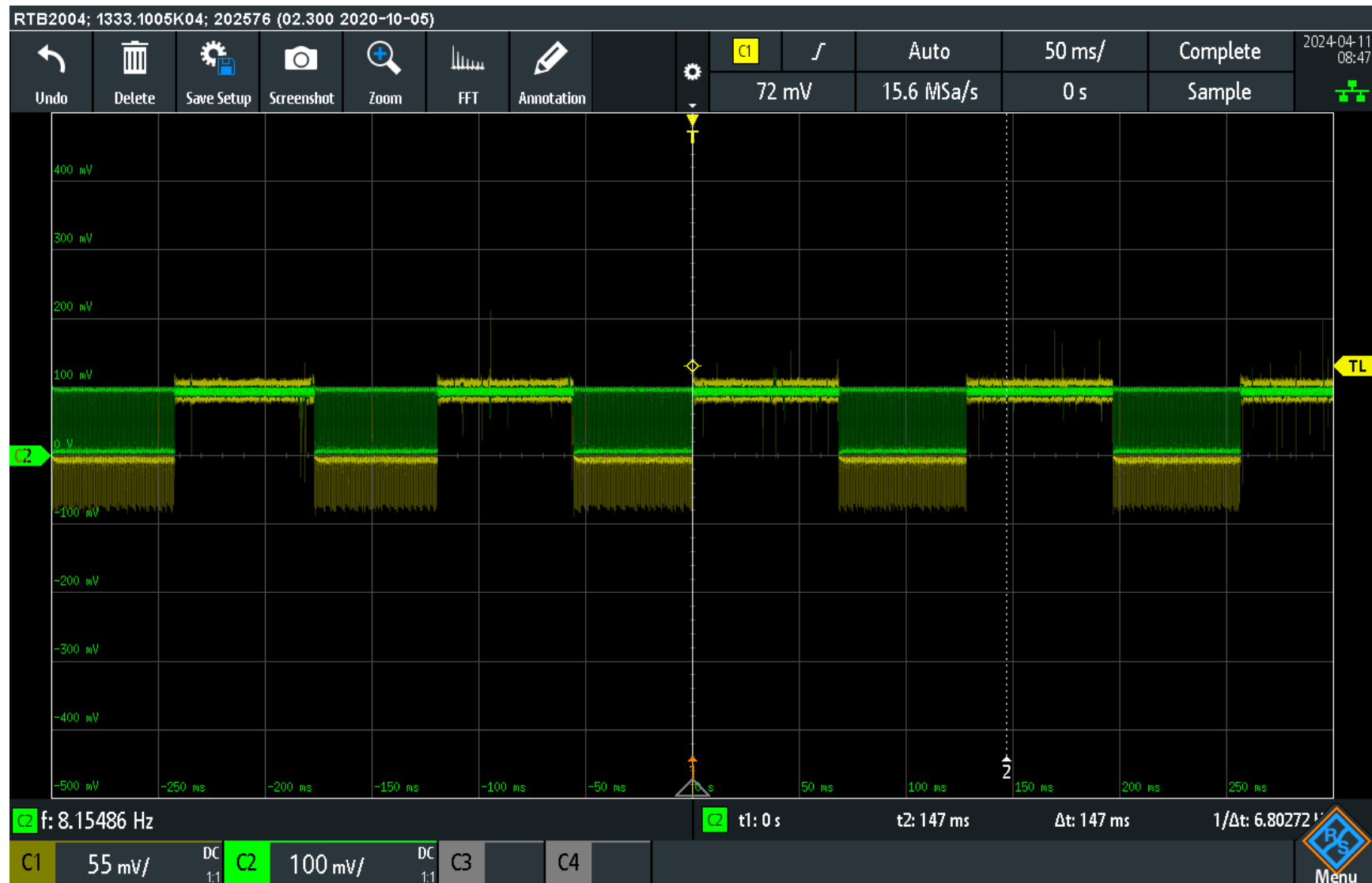


Figure 5: Experimental Results with the potentiometer set to ~66%



Figure 6: Experimental Results with the potentiometer set to 100%

Using these images we are able to determine two things

- i) The frequency output works as intended with fig.2 and fig.3 showing both low and high voltage outputs in a square output with a ~60% duty cycle and the expected frequency output of 242Hz -> 2424Hz.
- ii) The potentiometer intended to control the volume is also working as intended. Due to the logarithmic nature of the pot, each position of the dial corresponds to an appropriate volume output.

9. Appendix A

```
# Library script
#
# EAGLE Version 9.6.2 Copyright (c) 1988-2020 Autodesk, Inc.
#
Set Wire_Bend 2;
# Grid changed to 'mm' to avoid loss of precision!
Grid mm;
Layer 1 Top;
Layer 16 Bottom;
Layer 17 Pads;
Layer 18 Vias;
Layer 19 Unrouted;
Layer 20 Dimension;
Layer 21 tPlace;
Layer 22 bPlace;
Layer 23 tOrigins;
Layer 24 bOrigins;
Layer 25 tNames;
Layer 26 bNames;
Layer 27 tValues;
Layer 28 bValues;
Layer 29 tStop;
Layer 30 bStop;
Layer 31 tCream;
Layer 32 bCream;
Layer 33 tFinish;
Layer 34 bFinish;
Layer 35 tGlue;
Layer 36 bGlue;
Layer 37 tTest;
Layer 38 bTest;
Layer 39 tKeepout;
Layer 40 bKeepout;
Layer 41 tRestrict;
Layer 42 bRestrict;
Layer 43 vRestrict;
Layer 44 Drills;
Layer 45 Holes;
Layer 46 Milling;
```

Layer 47 Measures;
Layer 48 Document;
Layer 49 Reference;
Layer 51 tDocu;
Layer 52 bDocu;
Layer 88 SimResults;
Layer 89 SimProbes;
Layer 90 Modules;
Layer 91 Nets;
Layer 92 Busses;
Layer 93 Pins;
Layer 94 Symbols;
Layer 95 Names;
Layer 96 Values;
Layer 97 Info;
Layer 98 Guide;
Description ";

Edit 'RP8610-A10K-DUALGANG.sym';
Pin 'VIN1' io none middle R0 both 0 (12.7 10.16);
Pin 'OUT1' io none middle R0 both 0 (12.7 7.62);
Pin 'GND1' io none middle R0 both 0 (12.7 5.08);
Pin 'VIN2' io none middle R180 both 0 (40.64 10.16);
Pin 'GND2' io none middle R180 both 0 (40.64 5.08);
Pin 'OUT2' io none middle R180 both 0 (40.64 7.62);
Layer 94;
Change Style continuous;
Wire 0.254 (17.78 2.54) (17.78 12.7) (35.56 12.7) (35.56 2.54) \
(17.78 2.54);
Layer 95;
Change Size 1.778;
Change Ratio 8;
Change Align bottom left;
Change Font proportional;
Text '>NAME' R0 (17.78 15.24);
Layer 96;
Change Size 1.778;
Change Align bottom left;
Text '>VALUE' R0 (17.78 0);

Edit 'RP8756-A10K-DUALGANGPOT.pac';
Description ";
Layer 21;

```
Wire 0.127 (0 0) (9.5 0) (9.5 9.5) (0 9.5) \
(0 0);
Change Drill 1;Pad 'GND1' round 0 R0 (2.54 2.54);
Change Drill 1;Pad 'VOUT2' round 0 R0 (5.08 5.08);
Change Drill 1;Pad 'VOUT1' round 0 R0 (2.54 5.08);
Change Drill 1;Pad 'VIN1' round 0 R0 (2.54 7.62);
Change Drill 1;Pad 'VIN2' round 0 R0 (5.08 7.62);
Change Drill 1;Pad 'GND2' round 0 R0 (5.08 2.54);
Layer 25;
Change Size 1.27;
Change Align bottom left;
Text '>NAME' R0 (0 10.16);
Layer 27;
Change Size 1.27;
Change Align bottom left;
Text '>VALUE' R0 (0 -2.54);

Edit 'RP8756-A10K-DUALGANGPOT.dev';
Prefix 'POT';
Description ";
Value off;
Add RP8610-A10K-DUALGANG 'G$1' next 0 (-12.7 -5.08);
Package 'RP8756-A10K-DUALGANGPOT' ";
Technology ";
Connect 'G$1.GND1' 'GND1' 'G$1.GND2' 'GND2' 'G$1.OUT1' 'VOUT1' 'G$1.OUT2' 'VOUT2' 'G$1.VIN1' 'VIN1' \
'G$1.VIN2' 'VIN2';
Grid inch;
Edit ;
```

